전산화학 (2015년 1학기) **1차 시험 해답**

(총 6 문제, 60 점)

1. **(총 10점, 각 5점)**

(a) $z = a + bi$ (단, a와 b는 실수) 라면,

$$\begin{aligned} zz^* + 4(z - z^*) &= (a + bi)(a - bi) + 4(a + bi - a + bi) \\ &= a^2 + b^2 + 8bi = 5 + 16i \end{aligned} \tag{1}$$

복소수 상등정리를 이용하면 (1) 식으로부터 $a^2 + b^2 = 5$, 그리고 $8b = 16$. 그러므로 $b = 2$, $a = \pm 1$

i. $a = 1$, $b = 2$ 인 경우, $z = 1 + 2i$.

ii. $a = -1$, $b = 2$ 인 경우, $z = -1 + 2i$.

$\therefore z = \pm 1 + 2i$

(b) $z = a + bi$ (단, a와 b는 실수) 라면,

$$\begin{aligned} z^2 - 2z + 4 &= (a + bi)^2 - 2(a + bi) + 4 \\ &= a^2 + 2abi - b^2 - 2a - 2bi + 4 \\ &= a^2 - b^2 - 2a + 4 + (2ab - 2b)i = 0 \end{aligned} \tag{2}$$

복소수 상등정리를 이용하면 (2) 식으로부터

$$a^2 - 2a - b^2 + 4 = 0 \tag{3a}$$

$$2b(a - 1) = 0 \tag{3b}$$

(3b) 식으로부터 $a = 1$ 또는 $b = 0$ 이므로,

i. $a = 1$ 인 경우, (3a) 식으로부터 $b^2 = 3 \Rightarrow b = \pm\sqrt{3}$, 그러므로 $z = 1 \pm \sqrt{3}i$

ii. $b = 0$ 인 경우, (3a) 식으로부터 $a^2 - 2a + 4 = 0$. 이 식을 만족하는 실수 a를 구할 수 없다.

$\therefore z = 1 \pm \sqrt{3}i$

2. **(총 10점, 각 5점)**

(a)

$$\begin{aligned} z = 3e^{i\pi/4} + e^{i3\pi/2} &= 3\left(\cos\left(\frac{\pi}{4}\right) + i\sin\left(\frac{\pi}{4}\right)\right) + \cos\left(\frac{3\pi}{2}\right) + i\sin\left(\frac{3\pi}{2}\right) \\ &= 3\left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\right) - i \\ &= \frac{3\sqrt{2}}{2} + \left(\frac{3\sqrt{2}}{2} - 1\right)i \end{aligned}$$

$\therefore \mathrm{Re}(z) = \frac{3\sqrt{2}}{2}, \qquad \mathrm{Im}(z) = \frac{3\sqrt{2}}{2} - 1.$

(b) $2-\sqrt{12}i = 2-2\sqrt{3}i = 4\left(\frac{1}{2}-\frac{\sqrt{3}}{2}i\right) = 4\big(\cos(5\pi/3)+i\sin(5\pi/3)\big) = 4e^{i5\pi/3}$ 이므로

$$\begin{aligned} z=(2-\sqrt{12}i)^{28} &= \left(4e^{i5\pi/3}\right)^{28} \\ &= 4^{28}e^{i(28\times 5\pi/3)} \\ &= 2^{56}e^{i140\pi/3} \\ &= 2^{56}e^{i(46\pi+2\pi/3)} \\ &= 2^{56}e^{i2\pi/3} \\ &= 2^{56}\left(\cos\left(\frac{2\pi}{3}\right)+i\sin\left(\frac{2\pi}{3}\right)\right) \\ &= 2^{56}\left(-\frac{1}{2}+\frac{\sqrt{3}}{2}i\right) = -2^{55}+2^{55}\sqrt{3}i \end{aligned}$$

$\therefore \mathrm{Re}(z) = -2^{55}, \qquad \mathrm{Im}(z) = 2^{55}\sqrt{3}.$

3. **(총 10점, 각 5점)**

(a)

$$\begin{aligned} z=\frac{2i-1}{i-2} &= \frac{2i-1}{i-2}\times\frac{i+2}{i+2} \\ &= \frac{-2+4i-i-2}{-1-4} \\ &= \frac{4}{5}-\frac{3}{5}i \end{aligned}$$

$\therefore r = \sqrt{(4/5)^2+(-3/5)^2} = 1, \qquad \cos\theta = 4/5$ 이고, $\sin\theta = -3/5$ 이므로 $\theta = \cos^{-1}(4/5) = \sin^{-1}(-3/5)$

(b)

$$\begin{aligned} z=\frac{3i}{i-\sqrt{3}} &= \frac{3i}{i-\sqrt{3}}\times\frac{i+\sqrt{3}}{i+\sqrt{3}} \\ &= \frac{-3+3\sqrt{3}i}{-1-3} \\ &= \frac{3}{4}-\frac{3\sqrt{3}}{4}i \end{aligned}$$

$\therefore r = \sqrt{(3/4)^2+(-3\sqrt{3}/4)^2} = 3/2, \qquad \cos\theta = 1/2$ 이고, $\sin\theta = -\sqrt{3}/2$ 이므로 $\theta = -\pi/3+2n\pi$ (단, n은 정수)

4. **(10점)** de Moivre의 정리로부터

$$(\cos\theta+i\sin\theta)^3 = \cos 3\theta + i\sin 3\theta \tag{4}$$

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$$\begin{aligned} (\cos\theta+i\sin\theta)^3 &= \cos^3\theta+3\cos^2\theta(i\sin\theta)+3\cos\theta(i\sin\theta)^2+(i\sin\theta)^3 \\ &= \cos^3\theta+3i\cos^2\theta\sin\theta-3\cos\theta\sin^2\theta-i\sin^3\theta \\ &= \cos^3\theta-3\cos\theta\sin^2\theta+i(3\cos^2\theta\sin\theta-\sin^3\theta) \\ &= \cos^3\theta-3\cos\theta(1-\cos^2\theta)+i[3(1-\sin^2\theta)\sin\theta-\sin^3\theta] \\ &= 4\cos^3\theta-3\cos\theta+i(3\sin\theta-4\sin^3\theta) \end{aligned} \tag{5}$$

복소수 상등정리로부터 (4) 식과 (5) 식을 비교하면

$$\cos 3\theta = 4\cos^3\theta - 3\cos\theta,$$
$$\sin 3\theta = 3\sin\theta - 4\sin^3\theta$$

5. **(총 10점, 각 5점)**

(a)

$$x = r\cos\theta = 3 \times \cos\left(\frac{2\pi}{3}\right) = 3 \times -\frac{1}{2} = -\frac{3}{2}$$
$$y = r\sin\theta = 3 \times \sin\left(\frac{2\pi}{3}\right) = 3 \times \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2}$$

(b)

$$x = r\sin\theta\cos\phi = 2 \times \sin\left(-\frac{\pi}{6}\right) \times \cos\left(\frac{4\pi}{3}\right) = 2 \times -\frac{1}{2} \times -\frac{1}{2} = \frac{1}{2}$$
$$y = r\sin\theta\sin\phi = 2 \times \sin\left(-\frac{\pi}{6}\right) \times \sin\left(\frac{4\pi}{3}\right) = 2 \times -\frac{1}{2} \times -\frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}$$
$$z = r\cos\theta = 2 \times \cos\left(-\frac{\pi}{6}\right) = 2 \times \frac{\sqrt{3}}{2} = \sqrt{3}$$

6. **(총 20점)**

(a) **(5점)** Euler 공식으로부터

$$e^{i\theta} = \cos\theta + i\sin\theta, \tag{6a}$$
$$e^{-i\theta} = \cos\theta - i\sin\theta, \tag{6b}$$

그러므로 (6a) - (6b) 하면

$$\boxed{e^{i\theta} - e^{-i\theta} = 2i\sin\theta \Rightarrow \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}} \tag{7}$$

(b) **(5점)**

$$\begin{aligned}
\int_0^\pi e^{i2n\theta}d\theta &= \left[\frac{1}{2ni}e^{i2n\theta}\right]_0^\pi \qquad (n = \pm1, \pm2, \pm3, \cdots) \\
&= \frac{1}{2ni}\left(e^{i2n\pi} - 1\right) \\
&= \frac{1}{2ni}\left(\cos(2n\pi) + i\sin(2n\pi) - 1\right) \\
&\quad \cos(2n\pi) = 1 \quad \text{그리고} \quad \sin(2n\pi) = 0 \ \text{이므로} \\
&= 0
\end{aligned}$$

즉,

$$\boxed{\int_0^\pi e^{i2n\theta}d\theta = 0 \qquad (n = \pm1, \pm2, \pm3, \cdots)} \tag{8}$$

(c) **(10점)**

$$\begin{aligned}
\int_0^\pi \sin^4\theta d\theta &= \int_0^\pi \left(\frac{e^{i\theta} - e^{-i\theta}}{2i}\right)^4 \qquad (\text{(a)의 (7) 식으로부터}) \\
&= \frac{1}{16}\int_0^\pi (e^{i2\theta} - 2 + e^{-i2\theta})^2 d\theta \\
&= \frac{1}{16}\int_0^\pi (e^{i4\theta} + 4 + e^{-i4\theta} - 4e^{i2\theta} - 4e^{-i2\theta} + 2)d\theta
\end{aligned} \tag{9}$$

(b)의 (8) 식으로부터

$$\int_0^{\pi} e^{i4\theta} d\theta = \int_0^{\pi} e^{-i4\theta} d\theta = \int_0^{\pi} e^{i2\theta} d\theta = \int_0^{\pi} e^{-i2\theta} d\theta = 0 \tag{10}$$

0이므로 (9) 식은

$$\boxed{\int_0^{\pi} \sin^4\theta d\theta = \frac{1}{16}\int_0^{\pi} 6 d\theta = \frac{3}{8}\pi}$$